ZIMBABWE SCHOOL EXAMINATIONS COUNCIL (style)
ACADEX PRACTICE EXAMINATION
A-LEVEL · PURE MATHEMATICS

MATHEMATICS 6042/1
PAPER 1 November 2022
3 hours

These are **original ACADEX questions** written to match ZIMSEC syllabus structure, mark allocations and command words. They are **not** leaked or copied official papers.

Additional materials: A calculator may be used unless a question forbids it.
INSTRUCTIONS: Answer all questions. Full marks are not given for answers only — show method.

1. Find the first four terms in the expansion of $(1 + 2x)^5$ in ascending powers of x. **[8]**

2. The polynomial $p(x) = x^3 + ax^2 - 2x + 4$ has remainder 4 when divided by $(x - 1)$. Show this and find $p(-1)$. **[8]**

3. Differentiate $y = 5x^3 - 4x^2 + 2x - 1$. Hence find the gradient at $x = 1$. **[8]**

4. Find $\int (5x^2 - 4x + 2)\,dx$ and the definite integral from 0 to 1. **[8]**

5. Prove that $1 + \tan^2\theta = \sec^2\theta$. Hence solve $1 + \tan^2\theta = 4$ for $0° \le \theta \le 180°$. **[8]**

6. A is (1, 3), B is (7, 10).
(a) Midpoint [2]
(b) Distance AB [3]
(c) Equation of AB [3] **[8]**

7. An AP has first term 5 and common difference 3. Find (a) the 12th term [3] (b) S_{12} [5]. **[8]**

8. Solve $3^{(2x)} = 27$. Also simplify $\log_3 81$. **[8]**

9. $f(x) = 2x - 4$, $g(x) = x^2$.
(a) $fg(x)$ [2]
(b) $gf(x)$ [3]
(c) $f^{-1}(x)$ [3] **[8]**

10. Express $(5x+1)/((x+1)(x-2))$ in partial fractions. (Numbers may be checked by covering.) **[8]**

11. A = (4, 2), B = (7, 6). Find |AB| and a unit vector in the direction of AB. **[8]**

12. A circle has radius 6 cm. A sector has angle 2 radians.
(a) Arc length [3]
(b) Sector area [5] **[8]**

END OF QUESTION PAPER

Worked solutions and mark-scheme notes are inside the ACADEX app (Extract & Study).